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3 August 1964

MILITARY SPECIFICATION

GYROSCOPE, RATE SWITCHING CN-1050/ASN-55

1. SCOPE

1.1 This specification covers one type of rate switching gyroscope, designated CN-1050/ASN-55.

* 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

| | |
|---|---|
| QQ-P-416 | Plating, Cadmium (Electrodeposited) |
| PPP-B-636 | Box, Fiberboard |

Military

| | |
|---|---|
| MIL-P-116 | Preservation, Method of |
| MIL-E-5400 | Electronic Equipment, Aircraft, General Specification for |
| MIL-C-5541 | Chemical Films and Chemical Film Materials for Aluminum and Aluminum Alloys |
| MIL-S-7742 | Screw Threads, Standard, Optimum Selected Series: General Specification for |
| MIL-A-8625 | Anodic Coatings, for Aluminum and Aluminum Alloys |
| MIL-A-38329 | Attitude Heading Reference Set AN/ASN-55 |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-100 | Engineering Drawing Practices |
| MIL-STD-129 | Marking for Shipment and Storage |
| MIL-STD-130 | Identification Marking of U.S. Military Property |

FSC 6615

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| | |
|---|---|
| MIL-STD-143 | Specifications and Standards Order of Precedence for the Selection of |
| MIL-STD-454 | Standard General Requirements for Electronic Equipment |
| MIL-STD-461 | Electromagnetic Interference Characteristics Requirements for Equipment |
| MIL-STD-462 | Electromagnetic Interference Characteristics, Measurement of |
| MIL-STD-471 | Maintainability Demonstration |
| MIL-STD-781 | Reliability Tests, Exponential Distribution |
| MIL-STD-794 | Parts and Equipment, Procedures for Packaging and Packing of |
| MIL-STD-810 | Environmental Test Methods |
| MIL-STD-831 | Test Reports, Preparation of |
| MIL-STD-889 | Dissimilar Metals |
| MS3106 | Connector, Plug, Electric, Straight |

(Copies of documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

## 3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Components. The gyroscope shall consist of a control box containing a rate gyro and associated turn-switching circuits.

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials

3.4.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed inclosure, fungicidal treatments will not be necessary.

3.4.2 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

3.4.2.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion by means of protective coating or hermetic sealing, intimate contact of dissimilar metals shall be avoided. Dissimilar metals are defined by MIL-STD-889.



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3.4.3 Corrosive fumes. The materials, as installed in the gyroscope and under the service conditions specified herein, shall not liberate deleterious fumes.

3.4.4 Protective treatment. When materials are used in the contruction of the gyroscope that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating which will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.5 Design and construction. The gyroscope shall be designed to provide switching during turns. It shall be so constructed as to withstand the strains, jars, and vibrations incident to shipping and storage.

3.5.1 Switching mechanism. The gyroscope shall incorporate a rate gyro of a design suitable to provide the switching characteristics of the contacts as specified herein. The contacts shall be connected to pins D and E of the electrical connector. The circuit between pins D and E shall be normally open when power is disconnected or when there is zero rate of turn imposed. The time required for switching shall be as specified in table I. When a rate of turn of 15° per minute or more is imposed on the gyroscope, the circuit between pins D and E shall close. The contact rating shall be not less than 150 ma resistive load at 115V. The contact circuits shall be isolated from the other circuitry in the gyroscope and from the case.

TABLE I. Switching Requirements

| Rate of Turn (per min) | Room Temp | | Low Temp | | | High Temp |
|---|---|---|---|---|---|---|
| | Switching Time (sec) | Return to Normal (sec) | Switching (sec) -54°C | Return to Normal (sec) -54°C | Return to Normal (sec) -40°C | Return to Normal (sec) +71°C |
| 15° | 30 | 15 | 65 | 55 | 40 | 25 |
| 20° | 20 | 15 | 45 | 55 | 40 | 25 |
| 45° | 8 | 15 | 20 | 55 | 40 | 25 |
| 180° | 4 | 15 | 10 | 55 | 40 | 25 |



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3.5.2 Mounting orientation. The gyroscope shall be mounted with the mounting feet at the bottom and with the plane of the mounting feet perpendicular to the axis of the gyroscope about which it is turn-rate sensitive.

3.5.3 Hermetical seal. The complete gyroscope shall be hermetically sealed.

3.5.3.1 Filling medium. In the event the gyroscope is gas filled, the filling medium shall be 95 percent helium and 5 percent carbon dioxide. The gases used shall be at least 98 percent percent purity, free of dust particles, and shall contain not more than 0.006 milligram of water vapor per liter (dewpoint -65°C at filling pressure). The absolute pressure of the filling medium in the case shall be approximately 1 atmosphere.

3.5.3.2 Leak rate. The gyroscope shall be so sealed that the maximum leak rate will not permit more than 10 percent contamination of the filling medium after 1,000 hours at a pressure differential equivalent to 1 atmosphere.

* 3.5.4 Reliability. The gyroscope shall have a minimum acceptable mean-time-between-failures (MTBF) of 2,000 hours. The test program shall be based upon a discrimination ratio of 1.5 and a confidence level of 90 percent.

* 3.5.4.1 Longevity. The gyroscope shall have an operating life span (equipment longevity as defined in MIL-STD-781) of not less than 4,000 hours.

3.5.5 Maintainability. The following maintainability requirements shall be considered in the design of the gyroscope:

a. Minimization of complexity of maintenance tasks (such as calibration, adjustments, inspections, et cetera) by maximum use of simple design which includes optimum interchangeability and use of standardized equipment or commercial items

b. Rapid and positive recognition of equipment malfunction or marginal performance

c. Rapid and positive identification of the replaceable defective part, assembly, or component

d. Minimum numbers and types of tools and test equipment (special and standard) needed to perform maintenance

e. Optimum accessibility in all systems, equipments, and components requiring maintenance, inspections, removal or replacement.



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3.5.6 Compatibility. The gyroscope shall be so designed as to be compatible with the system requirements of the attitude heading reference set conforming to MIL-A-38329.

3.6 Performance. The gyroscope shall be capable of meeting the requirements specified herein when subjected to the following conditions:

a. Temperatures ranging from -65° to +71°C and up to 98°C for a period of 10 minutes

* b. Relative humidity up to 95 percent

c. Pressures ranging from 30 inches Hg down to 0.315 inch Hg, (approximately 100,000-foot altitude)

d. Exposure to salt fog for 48 hours

e. Vibrations up to 10g

f. Fungus enzymes as described in method 508 of MIL-STD-810 for a minimum of 28 days

g. Temperature shock from +85° to -40°C

h. Accelerations up to 20g.

* 3.6.1 Electromagnetic interference. Electromagnetic interference shall be within the limits specified in MIL-STD-461 for class ID equipment.

3.6.2 Yaw oscillation. The circuit between pins D and E shall remain open when the following oscillations are imposed:

a. ±1.0° with 2 second period

b. ±2.0° with 8 second period.

3.6.2.1 Steady-state turn and yaw oscillation. With the yaw defined in 3.6.2a superimposed on a steady 30° per minute turn, the switching time shall be not more than 30 seconds.

3.6.3 Dielectric strength. There shall be no breakdown of insulation when a potential of 400V dc through a 400-ohm resistor is applied between isolated pins and between pins and the case for 10-second periods.

3.6.4 Warmup. The gyroscope shall not require a warmup period of more than 2 minutes.



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3.7 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.8 Electronic parts. Electronic parts and the application thereof shall be in accordance with MIL-E-5400. Parts that do not appear on approved lists shall not be used unless approved by the procuring activity.

3.9 Electrical connector. A hermetically sealed connector shall be affixed to the gyroscope. The connector shall accept or mate with an MS3106C-14S-5S cable connector, or equivalent, and shall be equipped with a water-vaporproof cap and gasket. The pin connections shall conform to table II.

* TABLE II. Pin Connections and Circuit

| Pin | Circuit |
|---|---|
| A | 115V, 400-Hz, single-phase power |
| B | Ground |
| C | Reserved |
| D | Turn switching relay contact terminal (common) |
| E | Turn switching relay contact terminal (normally open) |

3.10 Power. The gyroscope shall operate from 115V, single-phase, 400-Hz power. The gyroscope shall function satisfactorily at variations in voltage and frequency from 103 to 127V and 380 to 420 Hz.

3.10.1 Power consumption. The total power consumption of the gyroscope shall not exceed 10 va when operating and shall not exceed 12 va during startup.

3.11 Weight. The weight of the gyroscope assembly shall not exceed 1.5 pounds.

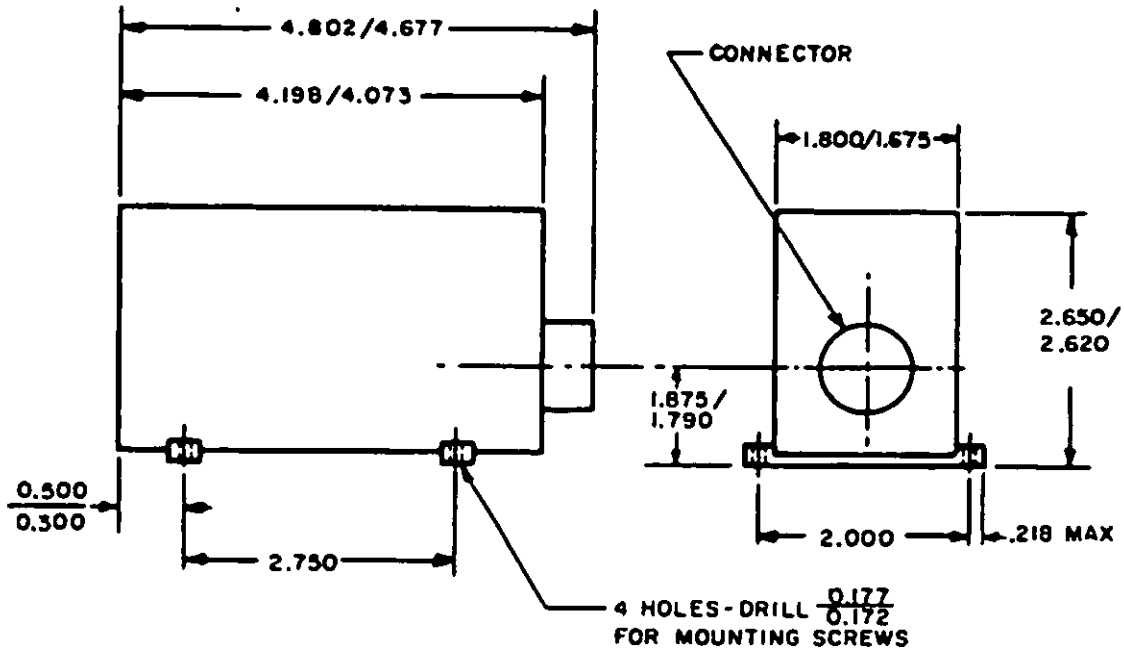
3.12 Dimensions. The outline dimensions of the gyroscope shall not exceed those shown on figure 1.

* 3.13 Soldering. Soldering shall be accomplished in accordance with MIL-STD-454.

3.14 Screw threads. Unless otherwise specified, the threads of all machine screws shall conform to MIL-S-7742.



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DIMENSIONS IN INCHES UNLESS
OTHERWISE SPECIFIED,
TOLERANCES : DECIMALS ± 0.010

* FIGURE 1. Outline Dimensions



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3.15 Finishes and protective coatings

3.15.1 Aluminum alloy parts. Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except as follows:

3.15.1.1 Small holes and case inserts need not be anodized.

3.15.1.2 Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

3.15.1.3 Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with MIL-C-5541 may be used in lieu of anodizing.

3.15.1.4 Coatings containing nonaluminum alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing.

3.15.1.5 When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

3.15.1.6 Parts enclosed in a hermetically sealed container need not be anodized or plated.

3.15.1.7 When the part is plated with tin over a copper flash, the part need not be anodized.

3.15.1.8 When necessary for electrical bonding, parts need not be anodized.

3.15.2 Steel parts. Where practicable, steel parts shall be cadmium plated in accordance with QQ-P-416, type II or III, as applicable, and of a class that is adequate to achieve the degree of protection required.

3.16 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.17 Workmanship. The gyroscope, including all parts and accesories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, brazing, painting, riveting, machine-screw assemblies, and freedom of parts from burrs and sharp edges.

3.17.1 Dimensions and tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the gyroscope, they shall be held or limited accordingly.



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3.17.2 Screw assemblies. Assembly screws and bolts shall be tight. The word tight means that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw or bolt or threads.

3.17.3 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the gyroscope shall be classified as follows:

a. Preproduction tests

b. Quality conformance tests.

4.3 Test conditions

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3.2 Attitude. Unless otherwise specified, the gyroscope shall be tested in its normal operating position.

4.3.3 Voltage input. Unless otherwise specified, the gyroscope shall be tested with 115V, single-phase, 400-Hz (nominal) power.

4.4 Preproduction testing

4.4.1 Test samples. The test samples shall consist of 18 gyroscopes representative of the production equipment. Three of the gyroscopes shall be subjected to all the tests except reliability and the other 15 gyroscopes shall be subjected to the reliability test only.



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4.4.1.1 Reliability demonstration. Unless otherwise specified, 15 gyroscopes representative of the production equipment on the contract or order shall be tested as specified in 4.6.15 for a total of 30,000 hours, unless rejection is determined in accordance with figure 2 prior to that time.

4.4.2 Test report, test samples, and data for the procuring activity. When the tests are conducted at a location other than the laboratory of the procuring activity, the following shall be furnished to that activity:

a. Test report - Three copies of a test report in accordance with MIL-STD-831

b. The samples which were tested and one untested sample

c. Two sets of assembly drawings

d. Brief operating instructions.

4.4.3 Preproduction tests. The preproduction tests shall consist of all the tests described under 4.6.

4.5 Quality conformance tests. Quality conformance tests shall consist of the following:

a. Individual tests

b. Sampling tests.

4.5.1 Individual tests. Each gyroscope shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Power consumption

c. Switching rate

d. Yaw oscillation

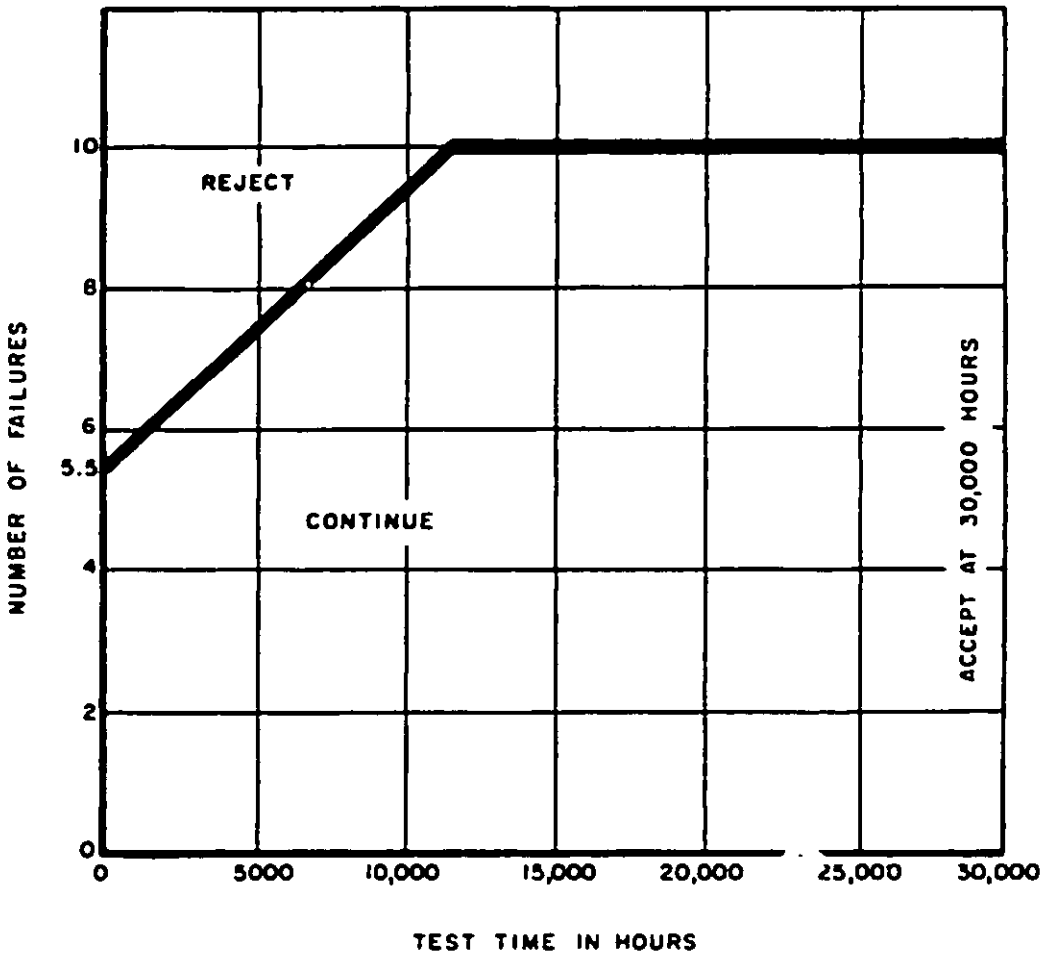
e. Steady state turn and yaw oscillation

f. Power failure

g. Leak rate.



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* FIGURE 2. Reject - Continue Test Criteria



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4.5.2 Sampling plans and tests

4.5.2.1 Sampling plan A. One gyroscope selected at random from each 100 or fraction thereof on the contract or order shall be subjected to the following tests as described under 4.6:

a. Individual tests

b. Low temperature operation

c. High temperature operation

d. Dielectric strength

e. Vibration error

4.5.2.2 Sampling plan B. Unless otherwise specified, 3 gyroscopes, selected at random from the first 15 items of the contract or order, shall be subjected to the following tests as described under 4.6:

a. Sampling plan A tests

b. High temperature exposure

c. High altitude - low temperature

d. Humidity

e. Fungus

f. Salt fog

g. Vibration failure

h. Acceleration

i. Temperature shock

j. Electromagnetic interference.

4.5.2.3 Rejection and retest. When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent of cause and failure are determined.



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4.5.2.3.1 Individual tests may continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or produced later shall not be made until it is determined that all items meet all the requirements of the specification.

* 4.5.2.4 Sampling plan C. Unless otherwise specified, 8 gyroscopes shall be selected at random from each 50 or less produced on the contract or order and tested as specified in 4.6.15, except 4.6.15.1, for a total of 13,200 hours.

* 4.5.2.4.1 Acceptance or rejection. Figure 3 shall be the criteria used to determine whether the equipment will be accepted or rejected. Certified time records shall be maintained of the test time accumulated on the equipment after starting the environmental cycling specified in 4.6.15. In the event a continue-test decision applies after 13,200 hours of testing, the production quantity shall be considered as conditionally acceptable; however, the accumulated time and failures shall be added to the data of the next production quantity for a composite accept-reject decision. If the continue test condition persists, the data from successive production quantities shall be accumulated until 10 failures or 30,000 hours of testing have occurred at which time an accept-reject decision shall be made.

4.5.2.4.2 Recording. Recording, data handling, and reporting procedures shall be in accordance with MIL-STD-781

4.5.2.5 Sampling plan D. One gyroscope shall be selected at random from each 25 produced on the contract or order and tested as specified in 4.6.17.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test methods

4.6.1 Examination of product. The gyroscope shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

4.6.2 Power consumption. Power consumption of the gyroscope shall not exceed 10 va during operation and shall not exceed 12 va during startup.

4.6.3 Switching rate. The gyroscope, with power applied and operating for not less than 1 minute nor more than 2 minutes, shall be mounted on a rate table and caused to turn at a rate of 15° per minute. At this rate of turn,



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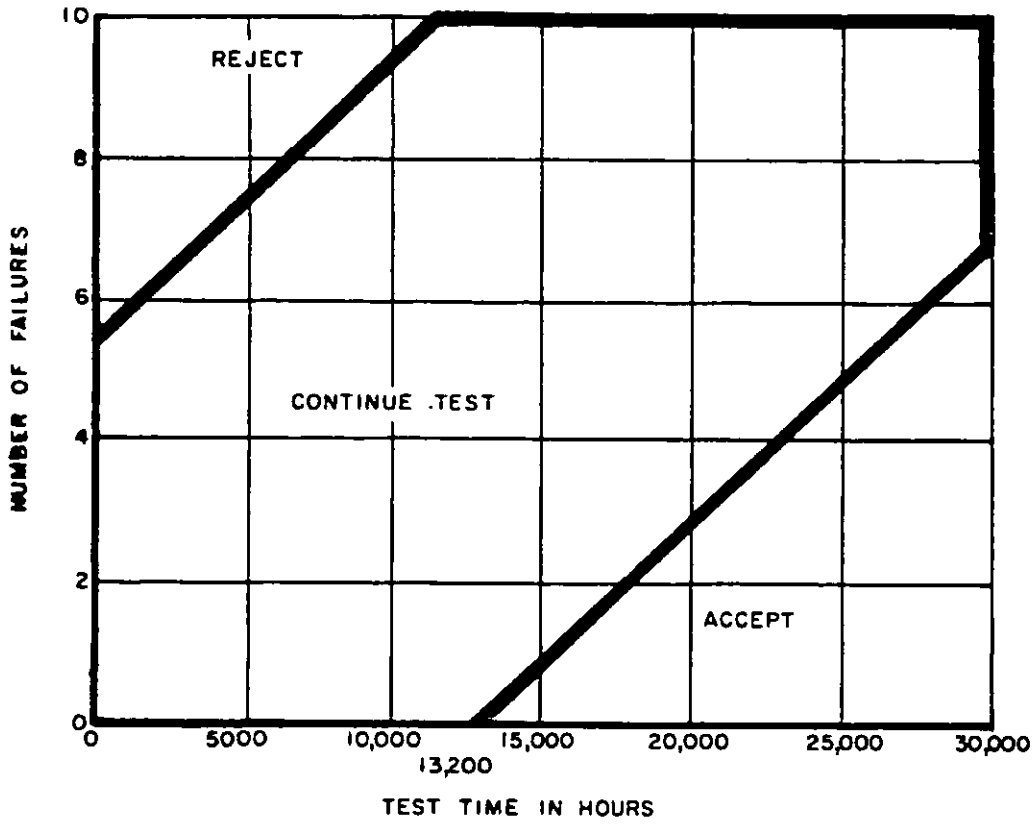



FIGURE 3. Reject - Continue Test - Accept Criteria



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the internal circuit between pins D and E shall close within 30 seconds. This procedure shall be repeated at 20° per minute, 45° per minute, and 180° per minute rates of turn. The same results shall occur within 20, 8, and 4 seconds, respectively. The switching rates shall be within the same limits regardless of the direction of turn. The faster rates may be tested first.

4.6.3.1 Return to normal. The circuit between pins D and E shall open within 15 seconds after each rate of turn specified in 4.6.3 has ceased.

4.6.4 Yaw oscillation. The gyroscope, with power applied and operating for not less than 1 minute, shall be mounted on a yaw test table. With zero steady-state turn imposed, the circuit between pins D and E shall remain open when the following oscillations in yaw are imposed:

a. $\pm 1.0°$ with 2-second period

b. $\pm 2.0°$ with 8-second period.

4.6.5 Steady state turn and yaw oscillation. The gyroscope, with power applied and operating for not less than 1 minute, shall be mounted on a test apparatus capable of superimposing a $\pm 1.0°$ with a 2-second period yaw oscillation on a 30° per minute rate of turn. These conditions shall be imposed on the gyroscope. The circuit between pins D and E shall close within 30 seconds and shall not return to normal until the steady-state turn has ceased. The results shall be the same regardless of direction of turn.

4.6.6 Power failure. The gyroscope, with power applied and operating for not less than 1 minute, shall be mounted on a rate table and turned at a rate sufficient to cause switching. After switching has occured, power to the gyroscope shall be removed. The circuit between pins D and E shall open immediately after removal of power.

4.6.7 Leak rate. The sealed case shall be tested for leaks by a helium leak detector or other means approved by the procuring activity. The leak rate shall not be sufficient to cause more than 10 percent loss or contamination of the filling medium after 1,000 hours at approximately 1 atmosphere differential pressure.

4.6.8 Low temperature operation. The gyroscope shall be properly connected and placed in a chamber at a temperature of $-54° \pm 2°C$ for a period of 4 hours. At the end of the 4-hour period and while the gyroscope is still at low temperature, power shall be applied. After power has been applied for 10 minutes, the gyroscope shall switch within 10, 20, and 45 seconds for rates of turn of 180°, 45°, and 20° per minute respectively, and shall meet the requirements specified in 4.6.6. After 15 minutes, the gyroscope shall switch within 65 seconds at a rate of turn of 15° per minute. In addition, after power has been applied for 15 minutes, switching contacts shall return to



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normal within 55 seconds while the gyroscope is at -54° ±2°C and within 40 seconds when it has been warmed to -40° ±2°C. The gyroscope may be vibrated at approximately 0.004 to 0.006 inch amplitude at a frequency of 10 to 12 Hz during this test.

4.6.9 High temperature operation. The gyroscope shall be properly connected and placed in a chamber at a temperature of 71° ±2°C for a period of 4 hours. The gyroscope shall then be removed from the 71°C ambient and immediately placed in a 98° ±2°C chamber for a period of 10 minutes after which it shall be immediately returned to a chamber at 71°C for a period of 30 minutes. The gyroscope shall be operated during this entire test. Immediately after the complete soak period and while the gyroscope is still at the 71° ±2°C temperature, it shall return to normal within 25 seconds after each turn rate is completed.

4.6.10 Dielectric strength. A potential of 400V dc through a 400-ohm resistor shall be applied between isolated pins and between pins and the case for a period of 10 seconds. There shall be no breakdown of insulation.

4.6.11 Vibration error. The vibration error test shall consist of a frequency survey with vibration applied to longitudinal, lateral, and vertical axes of the gyroscope. The gyroscope shall be subjected to vibration with a constant applied double amplitude of 0.060 inch through the frequency range of 10 to 55 Hz along the longitudinal and lateral axes of the case. The duration of the vibration shall be 1 minute in each axis for one complete cycle of frequency range. The vibration test equipment employed shall not apply a yaw rate greater than 30° per minute to the rate gyro. No switching shall occur.

4.6.12 Environmental. The following tests shall be conducted in accordance with the applicable methods of MIL-STD-810 and as specified:

4.6.12.1 High temperature exposure. The high temperature exposure test shall be conducted in accordance with method 501, procedure I, for a period of 24 hours, with the gyroscope properly connected but with no power applied. At the end of this period and while still at the high temperature, the gyroscope shall meet the test specified in 4.6.3. There shall be no sticking or erratic operation. The switching contacts shall return to normal within 25 seconds after each rate is completed.

4.6.12.2 Low temperature - high altitude. The gyroscope shall be properly connected, but with no power applied, and maintained at -65°C for 48 hours. Upon completion of this soak period, the temperature shall be maintained at -54°C for 24 hours. At the conclusion of the 72-hour soak period, the chamber pressure shall be reduced to the equivalent of 100,000 ±500 feet and power shall be supplied. The temperature shall be maintained at -54°C. After



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15 minutes of operation, the gyroscope shall be tested as specified in 4.6.3, 4.6.3.1, and 4.6.6. The time-to-switch rates shall not exceed the following:

| Rate of Turn (Per Min) | Switching Time (Sec) |
|---|---|
| 15° | 65 |
| 20° | 45 |
| 45° | 20 |
| 180° | 10 |

Return-to-normal time shall not exceed 55 seconds. The gyroscope may be vibrated at approximately 0.004 to 0.006 inch amplitude at a frequency of 10 to 12 Hz. After the gyroscope has returned to room temperature conditions, it shall meet all the individual tests.

4.6.12.3 Humidity. The humidity test shall be conducted in accordance with method 507, procedure I. After the humidity test cycle, the gyroscope shall meet the test specified in 4.6.3. There shall be no evidence of corrosion.

4.6.12.4 Fungus. The fungus test shall be conducted in accordance with method 508, procedure I. Upon completion of the test period, the gyroscope shall meet the test specified in 4.6.3. There shall be no deterioration nor shall any part of the gyroscope support fungus growth.

4.6.12.5 Salt fog. The salt fog test shall be conducted in accordance with method 509, procedure I, for a period of 48 hours. At the end of the 48-hour period, the gyroscope shall be subjected to the individual tests.

4.6.12.6 Vibration failure. The vibration failure test shall be conducted in accordance with method 514, equipment category (b), curve A, with power applied. At the end of this test, the gyroscope shall be subjected to the individual tests. No damage or irregular operation shall result from this test.

4.6.12.7 Temperature shock. The temperature shock test shall be conducted in accordance with method 503, procedure I. After the gyroscope has returned to room temperature, the connector and seals shall be examined. There shall be no evidence of cracked terminals or leaks in the seals. The gyroscope shall then meet the individual tests.

4.6.13 Acceleration. The gyroscope, not operating, shall be mounted on a centrifuge in its normal manner and subjected to an acceleration of 20g, first



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along its vertical axis and then along each of the two axes that are perpendicular to the vertical axis and to each other for a period of 1 minute in each axis. At the end of this acceleration, the gyroscope shall be subjected to the individual tests. No damage to the gyroscope shall result from this test.

4.6.14 Electromagnetic interference. The gyroscope shall meet the electromagnetic interference tests of MIL-STD-462 for class ID equipment when switching and while operating on a 15° total-movement roll, pitch, and yaw mechanism having 5 to 7 oscillations per minute.

4.6.15 Reliability. The gyroscope shall be properly connected with power applied as specified in 3.10 and subjected to the reliability test environment cycles specified on figure 4 while being operated on a 15° total-movement roll, pitch, and yaw mechanism having 5 to 7 oscillations per minute. After each 160 hours of operation, the gyroscope shall meet the individual tests.

* 4.6.15.1 Longevity. At the completion of the test specified in 4.6.15, three gyroscopes shall continue to be tested for a total of at least 4,000 hours each.

4.6.15.2 Relevancy criteria. All failures shall be classified as relevant unless approved for exclusion by the procuring activity.

4.6.15.3 Test information. Sixty days prior to starting this test, the contractor shall submit detailed test information for procuring activity approval. The information shall include but not necessarily be limited to:

a. Equipment used in monitoring performance

b. Periodicity of calibration checks to be made on monitoring equipment

c. Description of environmental test facilities to be used, including recording facilities, method of control of heating and cooling facilities, and vibration equipment

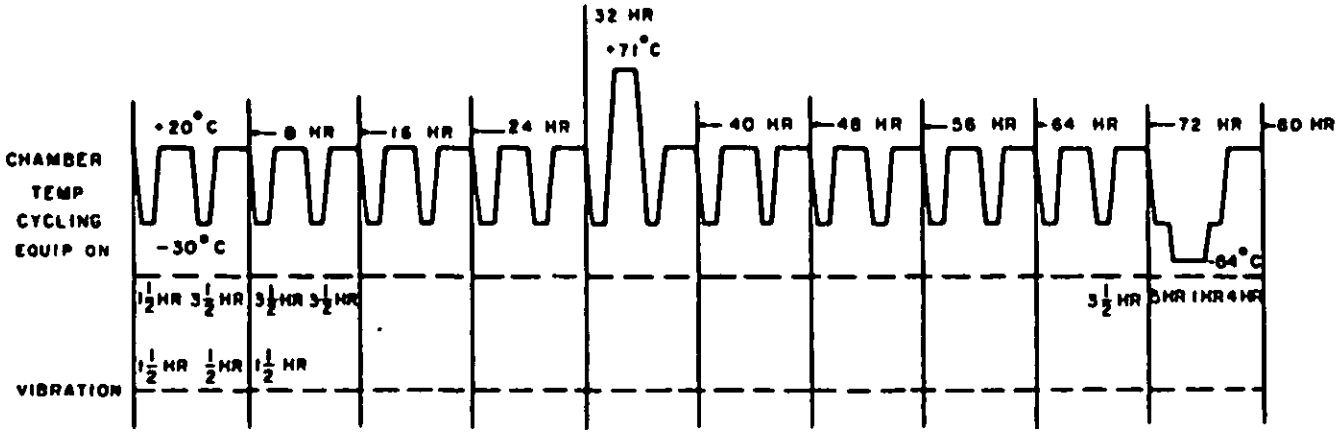
d. Frequency of monitoring.

* 4.6.15.4 Test records. Throughout this test, the contractor shall maintain adequate test records which shall be prepared in a form suitable for reproduction. As a minimum, the records shall include the following in accordance with MIL-STD-781: An operating sheet, log of failures, failure report, equipment repair sheet, failure tag, and failed-part summary analysis. Reproducible copies of the data shall be submitted to the procuring activity within 30 days after the completion of each test phase.

4.6.15.5 Repair of failed units. In the event of a failure during testing, the failed unit shall be repaired and the test continued.



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NOTES:

a. Environmental cycle repeats every 80 hours.
b. Temp rate of change of chamber shall be 0.1°C to 0.5°C per second.
c. Vibration level 0.2g at 20 to 30 Hz.
d. Off periods for equipment On and Off are 1/2 hour (except 76th hour).
e. Monitoring of system outputs for castastrophic-type failures performed during each ON cycle.
f. Satisfactory performance tests on systems and individual units conducted every 160 hours of environment cycle to detect performance-deviation-type failures.

FIGURE 4. Reliability Test Environment Cycles





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4.6.16 Maintainability demonstration. Maintainability demonstration shall be in accordance with MIL-STD-471, appendix B, method 2.

4.6.17 Compatibility. The gyroscope shall be tested for compatibility in accordance with MIL-A-38329.

4.7 Inspection of preservation, packaging, packing, and marking. The inspection of preservation, packaging, packing, and marking shall be in accordance with section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A. Each gyroscope shall be preserved in accordance with method III of MIL-P-116 and packaged in accordance with MIL-STD-794 in a container conforming to PPP-B-636.

5.1.2 Level C. Gyroscopes shall be provided sufficient protection against physical and mechanical damage to assure safe delivery, without degradation of item reliability, from the supply source to the first receiving activity for immediate use.

5.2 Packing

5.2.1 Level A. Gyroscopes packaged in accordance with 5.1.1 shall be packed in exterior-type shipping containers conforming to PPP-B-636, water-resistant grade.

5.2.2 Level B. Gyroscopes packaged in accordance with 5.1.1 shall be packed in exterior-type shipping containers conforming to PPP-B-636.

5.2.3 Level C. Packages that require overpacking for acceptance by the carrier shall be packed in exterior-type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall meet the consolidated freight classification rules or regulations of other common carriers as applicable to the mode of transportation.

5.3 Physical protection. Containerization, cushioning, blocking, bracing, bolting, and testing to determine adequacy of the completed package shall be in accordance with the applicable requirements of MIL-STD-794.

5.3.1 Verification. Rough handling tests of MIL-STD-794 for level A or C shall verify that the unit package will limit shock, vibration, or acceleration forces reaching the contained item to or below the values specified herein.



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5.4 Marking. Container marking shall be in accordance with the applicable requirements of MIL-STD-129. The nomenclature shall be as follows:

GYROSCOPE, RATE, SWITCHING CN-1050/ASN-55.

6. NOTES

6.1 Intended use. The CN-1050/ASN-55 gyroscope covered by this specification is intended for use in aircraft to sense turns above a specified rate and to switch erection circuits of a displacement gyroscope during these turns.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. When sampling plan B and C tests will not be conducted

c. Levels of preservation, packaging, and packing desired.

6.2.1 Failure report. During testing, a report of each failure that occurs should be submitted to the procuring activity within 2 working days after the failure.

6.3 Definitions

6.3.1 Hermetic seal. A hermetic seal is a perfectly closed and airtight seal made between vitric or metallic materials, or between vitric and metallic materials. A hermetic seal is not intended to include seals accomplished by gaskets.

6.3.2 Failure. Where a condition is specified as a requirement, a failure is considered as any electrical or mechanical condition that fails to comply with the requirements specified herein.

6.4 Marginal indicia. The outside margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in those notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 11

Preparing activity:
Air Force - 11

Review activity:
Air Force - 71

Project No. 6615-F009



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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